SNIST Sreenidhi Institute of Science and Technology AUTONOMOUS	**SREENIDHI INSTITUTE OF SCIENCE & TECHNOLOGY** (An Autonomous Institution) Yamnampet, Ghatkesar, Hyderabad – 501 301

ASSIGNMENT (UNIT- 4)

CLASS: B.Tech. II Year, II – Sem. **SUBJECT: PROBABILITY AND STATASTICS** **Branch: CSE,IT and EEE**

SHORT ANSWER TYPE QUESTIONS

1. Write the properties of t- distribution curve.
2. Write properties of Chi-square test?
3. Find $t_{0.025}$ when $\nu = 12.$
4. Find the value of $t_{0.995}$, when the size of the sample is 17.
5. Write the formula to calculated value of F-test.

LONG ANSWER TYPE QUESTIONS

1. Two horses A and B were tested according to the time (in seconds) to run on a particular track With the following results:

Horse A:	28	30	32	33	33	29	34
Horse B:	29	30	30	24	27	29	-

2.Test whether the two horses have the same running capacity. Use 0.05 LOS

The nicotine contents in mg in two samples of tobacco were found to be as follows:

Sample A:	24	27	26	21	25	
Sample B :	27	30	28	31	22	36

Can it be said that two samples have come from the populations have in same variances?

3. Test for goodness of Fit of a Poisson distribution at 0.05 L.O.S to the following distribution

Number of patients arriving/hour(x)	0	1	2	3	4	5	6	7	8
Frequency f :	52	151	130	102	45	12	5	1	2

4. From the following data, find whether there is any significant liking in the habit of taking soft drinks among the categories of employees.

	Clerks	Teachers	Officers
Pepsi	10	25	65
Thumpsup	15	30	65
Fanta	50	60	30

5. The average breaking strength of the steel rods is specified to be 18.5 thousand pounds. To test this sample of 14 rods was tested. The mean and S.D. obtained were 17.85 and 1.955 respectively. Is the result of experiment significant?

6. The heights of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% LOS assuming that for 9 dof.

7. In one sample of 8 observations the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in the other sample of 10 observations it was 102.6. Test whether this difference is significant at 5 % LOS.Pumpkins were grown under two experimental conditions. Two random samples of 11 and 9 pumpkins, show the sample S.D of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test hypothesis that the true variances are equal.

8. Fit a Poisson distribution and test the goodness of fit at 5% level

x:	0	1	2	3	4
f :	419	352	154	56	19

9. Four identical coins are tossed 160 times and the number of heads appearing each time is recorded as follows :

No. of Heads	0	1	2	3	4
Frequency :	14	30	70	35	11

Test the hypothesis that the coins are perfect at 5% level .

10. A certain drug was administrated to 500 people out of 800 included in a sample to test its efficiency against typhoid. The results are given below.

	Typhoid	No typhoid	Total
Drug	200	300	500
No Drug	280	20	300
Total	480	320	800

On the basis of the data can we say that the drug is effective in preventing typhoid?